





Quantum Teleportation (II)

Theory: How?

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What are we covering in this chapter?

- Quantum Teleportation: What and why?
- **Quantum Teleportation: How?** ✓
 - *Background: entanglement*
 - *Teleportation protocol*
 - *Teleportation: Theory example*
- QPiAI Explorer demonstration

Background: Entanglement

"Spooky action at a distance!"



- Entangled states: $|\Psi\rangle = \alpha|00\rangle + \beta|11\rangle$ $C = 0 \Rightarrow \text{No}$
- Measure of entanglement: $C = 2|C_{01}C_{10} - C_{00}C_{11}|$ $C = 1 \Rightarrow \text{Max.}$

$$|\Psi\rangle = \underline{C}_{00}|00\rangle + \underline{C}_{01}|01\rangle + \underline{C}_{10}|10\rangle + \underline{C}_{11}|11\rangle$$

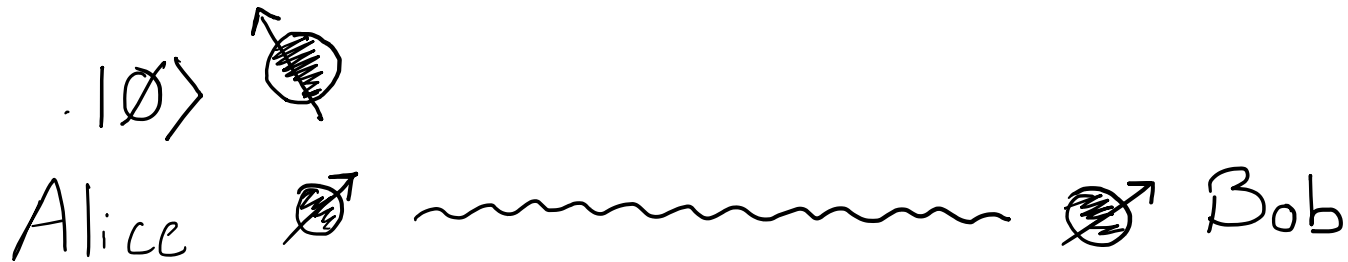
- Example: $|\Phi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$
 $|\Phi_2\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$
 $|\Phi_3\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Teleportation Protocol

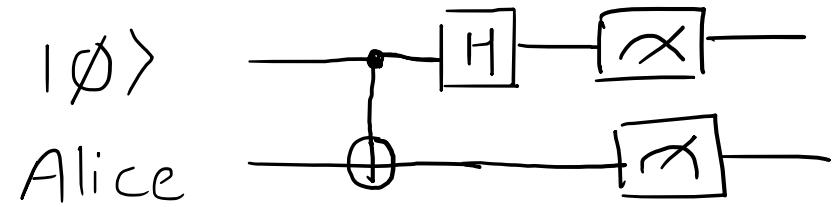
- Step 1: Share an entangled pair with Alice and Bob



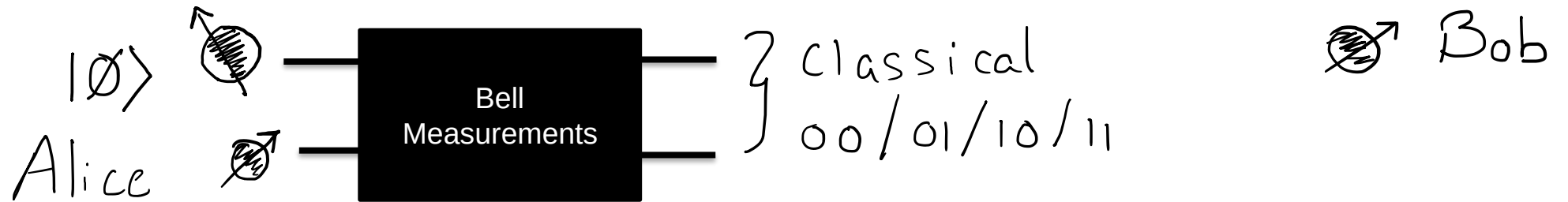
- Step 2: Let Alice prepare the state to be teleported



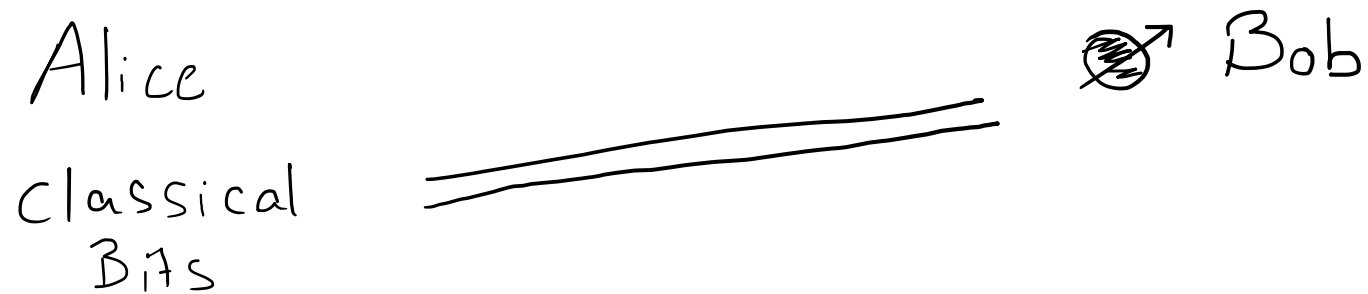
Teleportation Protocol



- Step 3: Perform bell measurements at Alice's end



- Step 4: Send the classical outcome to Bob

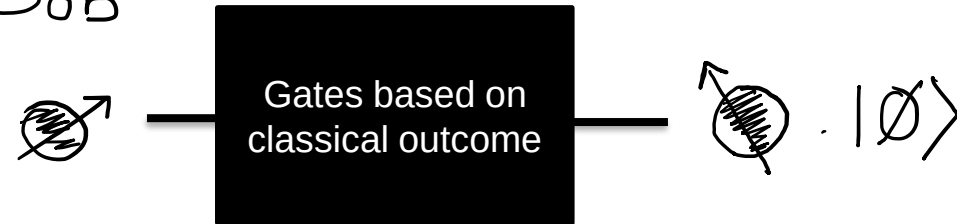


- Step 5: Bob performs I, X, Z or ZX operation, based on the classical bits received

Alice Classical Bits

Qubit $ \phi\rangle$	Alice's old qubit	
0	0	$\rightarrow$ I
0	1	$\rightarrow$ X
1	0	$\rightarrow$ Z
1	1	$\rightarrow$ ZX

Bob



Teleportation: Theory example

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle_{AB} + |11\rangle_{AB})$$

$$|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\phi\rangle \otimes |\psi\rangle = \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]$$

CNOT b/w 1st & 2nd & then H on 1st Qubit.

(i) CNOT b/w 1st & 2nd:

$\begin{matrix} 1 & 2 \\ 0 & 0 \end{matrix}$	$\rightarrow$	$\begin{matrix} 1 & 2 \\ 0 & 0 \end{matrix}$
$\begin{matrix} 0 & 1 \end{matrix}$	$\rightarrow$	$\begin{matrix} 0 & 1 \end{matrix}$
$\begin{matrix} 1 & 0 \end{matrix}$	$\rightarrow$	$\begin{matrix} 1 & 1 \end{matrix}$
$\begin{matrix} 1 & 1 \end{matrix}$	$\rightarrow$	$\begin{matrix} 1 & 0 \end{matrix}$

Teleportation: Theory example

$$(CNOT_{12} \otimes I)(|\phi\rangle \otimes |\psi\rangle) = \frac{1}{\sqrt{2}} [\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]$$

(ii) H on 1^{st} : $(H_1 \otimes I_2 \otimes I_3)(CNOT_{12} \otimes I)(|\phi\rangle \otimes |\psi\rangle)$

$$= \frac{1}{2} \left[\alpha(|000\rangle + |100\rangle + |011\rangle + |111\rangle) + \beta(|010\rangle - |110\rangle + |001\rangle - |101\rangle) \right]$$

$$= \frac{1}{2} \left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle) \right]$$

Teleportation: Theory example

Final state : $\frac{1}{2} \left[\begin{array}{l} |00\rangle (\alpha|0\rangle + \beta|1\rangle) \\ |01\rangle (\alpha|1\rangle + \beta|0\rangle) \\ |10\rangle (\alpha|0\rangle - \beta|1\rangle) \\ |11\rangle (\alpha|1\rangle - \beta|0\rangle) \end{array} \right]$

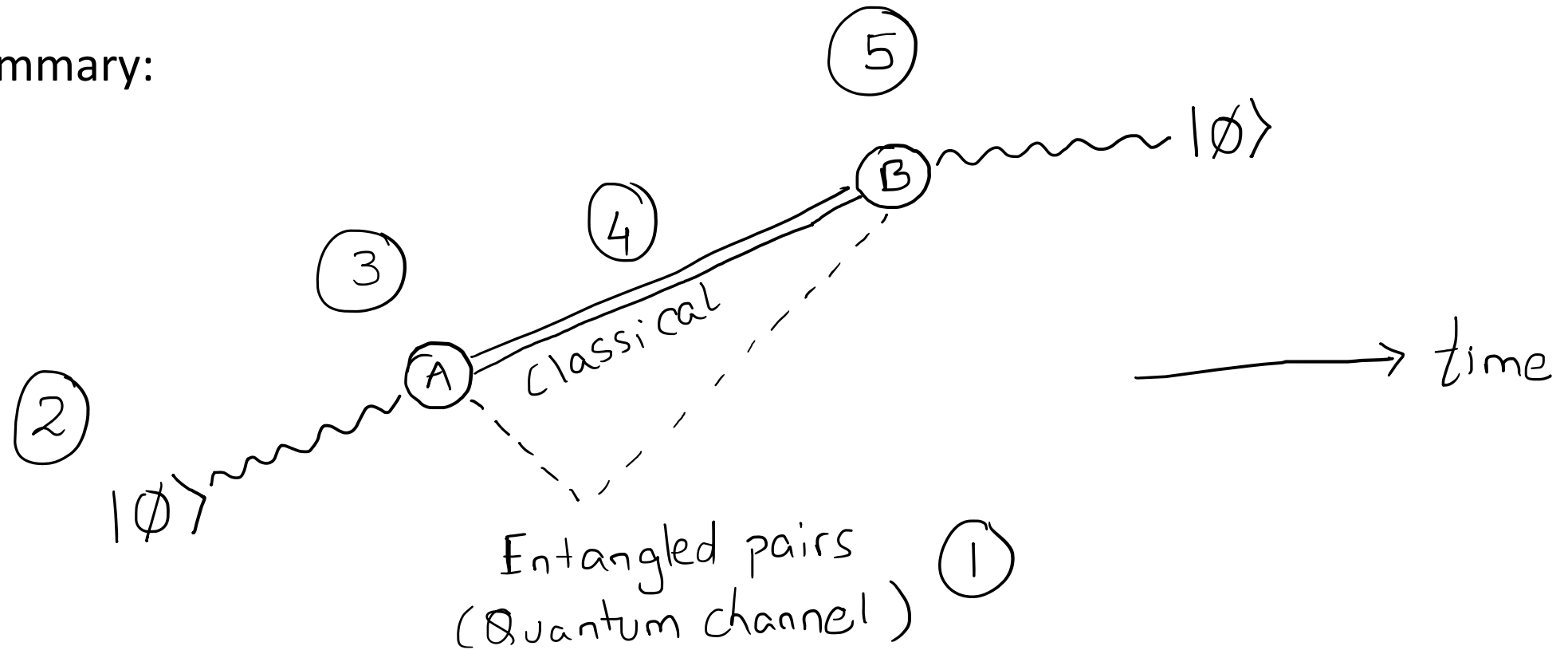
$\longrightarrow$ I
 $\longrightarrow$ X
 $\longrightarrow$ Z
 $\longrightarrow$ ZX

Qubit $|\phi\rangle$ $\swarrow$
 Alice's old Qubit $\searrow$

Bob's Qubit.

Teleportation Protocol

- Summary:





Thank you for your attention!

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